

Georeferenced 360-Degree Photos for Enhancing Navigation and Interaction within Virtual Electric Power Substations

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ABSTRACT

This work introduces a methodology for improving virtual navigation and interaction within power substations by employing georeferenced 360-degree photos. The objective is to swiftly update the current state of the field since, in most cases, incorporating such amendments is not feasible during the reconstruction of virtual environments. This changing speed is very important for critical systems. Preliminary results demonstrate successful updating rates, enabling engineers to make rapid decisions. It is expected that the proposed methodology can improve the efficiency and dependability of step-by-step activities, while also reducing the time and costs associated with system maintenance.

Keywords: Virtual Reality, 360-degree photos, geospatial technologies, electric power substations, system safety and maintenance.

Index Terms: Human-centered computing–Human computer interaction (HCI)–Interaction paradigms–Virtual reality; Hardware–Power and energy–Energy distribution

1 INTRODUCTION

Virtual Reality (VR) systems present a transformative leap in user interfaces, delivering an immersive three-dimensional experience that transcends the constraints of traditional two-dimensional layouts [1]. This heightened sense of immersion promotes the development of a mental model closely aligned with real-world field operations [2]. VR has emerged as a pivotal tool in overcoming the limitations of 2D interaction spaces, enabling users to engage with information in a manner that mirrors the complexities of actual physical environments [3]. At the heart of this experience is the feeling of presence, a key feature of VR environments that significantly enhances user engagement [3].

In the power generation industry, where security is a top priority, optimizing processes and logistics is equally crucial for cost reduction. When issues arise within a station, power plant, or transmission network, prompt resolution requires a skilled team [2]. However, the recruitment of experienced personnel presents challenges in terms of both availability and cost-effectiveness [4]. Managing and maintaining critical infrastructures, especially electric power substations, demands efficient and reliable approaches. Moreover, the swift update of the current field status and the integration of such changes are impractical during the

reconstruction of virtual environments [2], [4]. This rapid rate of change holds significant importance for critical systems.

In this context, georeferenced 360-degree photos stand out as a valuable asset, offering a unique set of advantages for navigation and interaction within virtual electrical power substations [5]. By integrating these high-resolution images into a 3D virtual environment, users can explore and interact with equipment in a way that closely mirrors real-world conditions.

The rapid evolution of geospatial technologies has paved the way for innovative solutions in the management and interaction with such infrastructures. This work introduces a methodology that harnesses the power of geospatial technologies for virtual navigation and equipment interaction within substations. The objective of this research is to swiftly update the current state of the field, recognizing that in most cases, incorporating such amendments is not feasible during the reconstruction of virtual environments. By incorporating georeferenced 360-degree photos into the virtual environment of a substation in VR, the proposed methodology can ensure a safe and controlled environment for training and testing. Simultaneously, it can increase the efficiency and reliability of step-by-step activities, thereby reducing the time and costs associated with system maintenance [6]. This approach addresses the multifaceted challenges of the power generation industry, offering a promising path for advancing safety, efficiency, and overall operational excellence in electrical power substations.

2 PROPOSED METHODOLOGY

The proposed methodology involves a comprehensive approach from capturing images in the field to utilizing the virtual environment, as depicted in Figure 1. Initially, field operators use a 360-degree camera to capture detailed images of the substation, also recording information about each equipment during the process.

The captured images undergo an extraction process of Exchangeable Image File Format (EXIF) information, including crucial data such as date, time, GPS location, and camera orientation. These details serve as the foundation for georeferencing the images, ensuring the precise location of equipment in the substation. Image processing and pattern recognition techniques are applied to identify specific objects in the images, such as equipment tags.

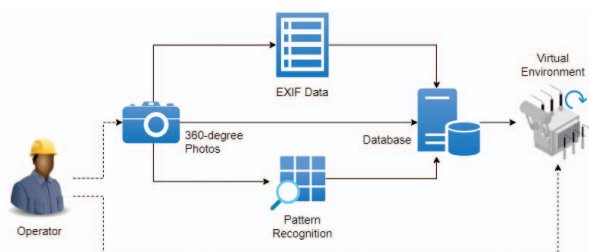


Figure 1: Diagram of the proposed methodology.

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The extracted data and operator records are stored in a centralized database, allowing easy updating and insertion of new information as new images are captured. This data supports the 3D modeling of substation components. Furthermore, based on this data, models are integrated into the virtual environment and positioned according to geospatial coordinates in the substation.

The insertion of captured photos into the virtual environment occurs in an overlaid manner, corresponding to the positions of the 3D models of the equipment, providing a precise and contextualized visualization. This virtual environment is continuously updated whenever new images are captured, ensuring it accurately reflects changes in the substation over time.

The resulting system is accessed by operators, engineers, and maintenance teams for asset management, safety training, and operational planning. Interactive features are provided to facilitate analysis and decision-making, promoting an effective approach that prioritizes safety, efficiency, and innovation in the electric power sector.

3 PRELIMINARY RESULTS

In the initial stages of our research, we applied the proposed methodology to the *Mascarenhas de Moraes Hydroelectric Power Plant* substation operated by *Eletrobras Furnas*. At this developmental phase, the substation is being modeled utilizing *Autodesk Inventor*, while the corresponding virtual environment is constructed using *Unity3D* to incorporate interactive features and navigation capabilities.

A total of 550 high-resolution 360-degree photos were captured from various vantage points within the substation. The EXIF information extracted from these photos was then integrated into our centralized database, ensuring a robust foundation for subsequent georeferencing processes.

Implementation of the proposed methodology facilitated the insertion of these georeferenced photos into the virtual environment. Each photo was represented by a virtual sphere positioned near the respective equipment within the virtual substation. This dynamic integration of georeferenced 360-degree photos allowed for a contextualized visualization within the virtual space.

In our preliminary exploration of the system, users were able to navigate through the virtual environment effortlessly. Upon approaching a virtual sphere and executing a click, the corresponding 360-degree photo opened in the same spatial location within the virtual environment. This functionality enabled users to seamlessly transition between the virtual and real environments, offering an immersive and location-specific view of the substation, as shown in Figure 2.



Figure 2: Virtual substation with inserted spheres (left) and a 360-degree photo opened in a specific position (right).

As a crucial element, the continuous update of the virtual environment was tested, involving the addition of new photos to the database while leveraging EXIF date and time information. The successful update showcased the system's rapid integration rates, providing engineers with new insights as soon as photos are updated, since the same for reconstructing the VR scene would take longer.

The pattern recognition algorithm demonstrated the capability to recognize equipment tags and place corresponding photos in specific locations. However, refinement is needed as the development is ongoing. Future iterations will focus on addressing challenges to enhance the overall effectiveness of the methodology.

4 CONCLUSION

This work introduced a methodology for virtual navigation and interaction within power substations, employing georeferenced 360-degree photos in a 3D virtual environment. Preliminary results from applying this methodology to the *Mascarenhas de Moraes Hydroelectric Power Plant* substation demonstrated successful photo capture, georeferencing, and integration into the Virtual Reality space. Users experienced an immersive virtual environment, allowing detailed navigation and location-specific insights through interactive 360-degree photos.

Updating rates, which involves incorporating new photos into the database, has demonstrated satisfactory for engineering decisions, facilitating quick responses to accidents and unforeseen events in substations. This capability enhances practical applicability by providing immediate insights with new photo additions. While the integration of georeferenced photos is a significant achievement, ongoing development focuses on refining the pattern recognition algorithm.

Looking forward, the proposed methodology holds potential to help the power industry manage and maintain critical infrastructures. By combining georeferenced photos with virtual navigation, our approach aims to enhance safety, efficiency, and innovation in the electric power sector.

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